MERN STACK

DAY – 29

28 July 2026

PROJECT: PARKEASY - PARKING MANAGEMENT SYSTEM

USER SCHEMA DEVELOPMENT

The User Schema is created using Mongoose to define the structure of user data stored in the MongoDB database.

• It is used to store user information required for authentication and parking management operations.

• The User Schema is connected with the User model using:

```
mongoose.model("User", userSchema)
```

• The User Schema contains the following fields:

1. **Name** : Stores the user's full name.
2. **Email**
    - Stores the user's email address.
    - The email field is unique to prevent duplicate accounts.
3. **Password** : Stores the user's password for authentication.
4. **Role** : Stores the user type such as:
    - User
    - Admin
5. **Vehicle Details** : Stores vehicle information of the user.

    Vehicle contains:
    - Vehicle Number
    - Vehicle Type

6. **Bookings Array** : Stores parking booking details of users.

   Booking information includes:

   - Slot Number
   - Vehicle
   - Booking Date
   - Start Time
   - End Time
   - Booking Status
   - Entry Time
   - Exit Time
   - Parking Status
   - Parking Duration

• Default values are provided for fields like booking status, entry time, exit time, and parking status.
• The schema helps maintain a structured format for storing user-related data in MongoDB.

Working

1. The mongoose.Schema() function is used to create the structure of user documents.
2. User details such as name, email, password, and role are defined inside the schema.
3. Vehicle information is stored as an embedded object inside the user document.
4. Parking bookings are stored as an array of objects inside each user document.
5. The User model is created using the defined schema.

6. The User model is used in signup and login APIs to store and retrieve user information from MongoDB.